Stochastic programming
Lagrangian methods

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Motivation

Reference: Birge and Louveaux, Sections 3.4, 5.8 and 6.4.

We consider the general two-stage nonlinear program

$$\begin{aligned} \inf z &= f^1(x) + \mathcal{Q}(x) \\ \text{s.t. } & g_i^1(x) \leq 0, \ i = 1, \ldots, m_1, \end{aligned}$$

where $\mathcal{Q}(x) = \mathbb{E}_{\boldsymbol{\xi}}[Q(x, \boldsymbol{\xi})]$, and

$$\begin{aligned} Q(x, \xi) = \inf \ & f^2(y(\xi), \xi) \\ \text{s.t. } & b_i^2(x, \xi) + g_i^2(y(\xi), \xi) \leq 0, \ i = 1, \ldots, m_2. \end{aligned}$$

We assume that all functions $f^2(\cdot, \xi)$, $b_i^2(\cdot, \xi)$ and $g_i^2(\cdot, \xi)$ are continuous in the first argument for any given ξ, and measurable in ξ for any fixed first argument. $\boldsymbol{\xi}$ has finite second-order moments.

Properties - assumptions

We extend the definitions of K_1 and K_2 of the linear case (two-stage slides).

$$
\begin{aligned}
K_1 &= \{x \mid g_i^1(x) \leq 0,\ i = 1, \ldots, m_1\}, \\
K_2(\xi) &= \{x \mid \exists y(\xi) \text{ s.t. } b_i^2(x, \xi) + g_i^2(y(\xi), \xi) \leq 0,\ i = 1, \ldots, m_2\}, \\
K_2 &= \{x \mid \mathcal{Q}(x) < \infty\}.
\end{aligned}
$$

Assumptions.

1. Convexity. The function f^1 is convex on $\mathbb{R}^{n_1}$, g_i^1 is convex on $\mathbb{R}^{n_1}$ $(i = 1, \ldots, m_1)$, $f^2(\cdot, \xi)$ is convex on $\mathbb{R}^{n_2}$ for all $\xi \in \Xi$, $g_i^2(\cdot, \xi)$ is convex on $\mathbb{R}^{n_2}$ $(i = 1, \ldots, m_2)$, for all $\xi \in \Xi$, and $b_i^2(\cdot, \xi)$ is convex on $\mathbb{R}^{n_1}$ $(i = 1, \ldots, m_2)$, for all $\xi \in \Xi$.
2. Slater condition. If $\mathcal{Q}(x) < \infty$, for almost every (a.e.) $\xi \in \Xi$, $\exists$ some $y(\xi)$ s.t. $b_i^2(x, \xi) + g_i^2(y(\xi), \xi) < 0$ $(i = 1, \ldots, m_2)$.

Second-stage properties

The Slater condition ensures that the *strong duality* holds at the second stage and the KKT conditions are necessary and sufficient.

Theorem

Under assumptions 1 and 2, the recourse function $Q(x, \xi)$ is a convex function in x for all $\xi \in \Xi$.

Theorem

If the recourse feasible set (i.e. in y, for given ξ) is bounded for any $x \in \mathbb{R}^{n_1}$, then the function $Q(x, \xi)$ is lower semi-continuous at x for all $\xi \in \Xi$.

Lower semi-continuity

We say that a function f is lower semi-continuous at x_0 if for every $\varepsilon > 0$, $\exists$ a neighborhood U of x_0 such that $f(x) > f(x_0) - \varepsilon$ for all $x \in U$. Equivalently

$$\liminf_{x \to x_0} f(x) \geq f(x_0),$$

where

$$\liminf_{x \to a} f(x) = \lim_{\varepsilon \to 0} \inf\{f(x) : x \in \mathrm{dom}(f) \cap B(a; \varepsilon) \backslash \{a\}\}.$$

The function f is lower semi-continuous if it is lower semi-continuous at every point.

Lower semi-continuity (cont'd)

A function is lower semi-continuous iff $\{x \in X : f(x) > \alpha\}$ is an open set for every $\alpha \in \mathbb{R}$, or, in a similar way, $\{x \in X : f(x) \leq \alpha\}$ is a closed set for every $\alpha \in \mathbb{R}$.

Example of lower semi-continuous function

$$f(x) = \begin{cases} x & \text{if } x \leq 1 \\ x + 1 & \text{if } x > 1 \end{cases}$$

Convexity (cont'd)

Corollary

The expected recourse function $\mathcal{Q}(x)$ is a convex function in x.

Corollary

The feasible set $K_2 = \{x \mid \mathcal{Q}(x) < \infty\}$ is convex, and closed under the assumptions of Birge and Louveaux, Section 3.4.

Convexity is inherited from $\mathcal{Q}$, closedness is not: the domain of a convex — even lower semi-continuous — function need not be closed, as $x \mapsto 1/x$ on $(0, \infty)$ shows. It comes from the continuity of the second-stage functions when the recourse feasible sets are bounded: if $x_n \to \bar{x}$ with feasible y_n, the y_n stay bounded (by convexity of the g_i^2), and a limit point of (y_n) is feasible for $\bar{x}$. Each $K_2(\xi)$ is then closed, and so is K_2 when Ξ is finite.

Corollary

If the recourse feasible set is bounded for any $x \in \mathbb{R}^{n_1}$, $\mathcal{Q}$ is a lower semi-continuous function.

Solution: existence

Theorem

If the recourse feasible set is bounded for any $x \in \mathbb{R}^{n_1}$, K_1 is bounded, f^1 is continuous, g_i^1 and g_i^2 are continuous for every i, and $K_1 \cap K_2 \neq \emptyset$, then the nonlinear stochastic two-stage program has an optimal solution and the infimum is reached.

Relative interior

The relative interior of a set S, denoted by $ri(S)$, is defined as its interior within the affine hull of S. In other words

$$ri(S) = \{x \in S \,|\, \exists\, \varepsilon > 0, (B_\varepsilon(x) \cap \mathrm{aff}(S)) \subseteq S\},$$

where $\mathrm{aff}(S)$ is the affine hull of S, and $B_\varepsilon(x)$ is a ball of radius ε centered at x. Any norm can be used for the ball: in finite dimension, all norms define the same relative interior.

The affine hull $\mathrm{aff}(S)$ of S is the set of all affine combinations of elements of S, i.e.

$$\mathrm{aff}(S) = \left\{ \sum_{i=1}^{k} \alpha_i x_i \,\middle|\, x_i \in S,\ \alpha_i \in \mathbb{R},\ \sum_{i=1}^{k} \alpha_i = 1, k = 1, 2, \dots \right\}.$$

Solution: optimality

Theorem

Assume that the Slater condition is satisfied:

$$\exists\, x \text{ s.t. } x \in ri(dom\, f^1),\ x \in ri(dom\, \mathcal{Q}) \text{ and } g_i^1(x) < 0,\ i = 1, \ldots, m_1.$$

x^* is optimal iff there exists λ^* such that (x^*, λ^*) satisfies the KKT conditions for the two-stage stochastic program, i.e.

- $x^* \in K_1$,
- $\lambda^* \geq 0$,
- $\lambda_i^* g_i^1(x^*) = 0,\ i = 1, \ldots, m_1$,
- $0 \in \partial f^1(x^*) + \partial \mathcal{Q}(x^*) + \sum_{i=1}^{m_1} \lambda_i^* \partial g_i^1(x^*)$.

Basic principle of Lagrangian approach

- Nonlinear problem.
- We aim to build a search direction and compute a step along this direction to reduce the objective.
- Problem: classical nonlinear methods assume that the (sub-)gradients of $\mathcal{Q}$ are available and cheap to obtain. Not the case here.
- Link the first and second stages in order to avoid optimization subproblems when building search directions.

Basic principle of Lagrangian approach (cont'd)

- Let π be a vector of (dual) multipliers associated to the second-stage constraints. (Notation: π carries the second-stage and nonanticipativity multipliers throughout the course, λ those of the first-stage constraints g_i^1, as above.)
- Dual problem with respect to the second-stage constraints (linking the first and second stages):

$$\max_{\pi(\xi)\geq 0} w = \theta(\pi),$$

where

$$\theta(\pi) = \inf_{x,y} f^1(x) + \mathbb{E}_\xi[f^2(y(\xi),\xi)] + \mathbb{E}_\xi\left[\sum_{i=1}^{m_2} \pi_i(\xi)\left(b_i^2(x,\xi) + g_i^2(y(\xi),\xi)\right)\right]$$
$$\text{s.t. } g_i^1(x) \leq 0,\ i = 1,\ldots,m_1.$$

Duality

Theorem

Assume that

- *all the functions of the stochastic program are convex,*
- *there exists a finite optimal value,*
- *the Slater constraint qualification holds,*
- *the support of $\boldsymbol{\xi}$ is finite, $\Xi = \{\xi_1, \ldots, \xi_S\}$, with probabilities $p_1, \ldots, p_S$.*

Then $z \geq w$ for all $x, y_1, \ldots, y_S$ feasible in the primal formulation and $\pi_1, \ldots, \pi_S$ feasible in the dual formulation (weak duality).

Moreover, their optimal values coincide: $z^* = w^*$ (strong duality).

We thus assume from now on that Ξ is finite, and we will describe some procedures exploiting the Lagrangian function.

Basic Lagrangian dual ascent method

Assumption: the minimization that defines $\theta(\pi)$ has a unique minimizer.

Algorithm 1: Lagrangian dual ascent method

Step 0. Let $\pi^{0,s} \geq 0$, $s = 1, \ldots, S$, $\pi^0 = (\pi^{0,1}, \ldots, \pi^{0,S})$, $\nu = 0$.

Step 1. Given $\pi = \pi^\nu$ in the dual problem, consider the solution $(x^\nu, y_1^\nu, \ldots, y_S^\nu)$. For $s = 1, \ldots, S$, define

$$r_i^s = b_i^2(x^\nu, \xi_s) + g_i^2(y_s^\nu, \xi_s),$$

the residual of the second-stage constraints. Stop when $r_i^s \leq 0$ and $\pi_i^{\nu,s} r_i^s = 0$ for every i, s: feasibility and complementarity.

Step 2. Choose a step $\alpha^\nu \geq 0$ and set $\pi^{\nu+1,s} = \max\{\pi^{\nu,s} + \alpha^\nu r^s, 0\}$. Set $\nu := \nu + 1$ and check convergence. If we have not yet converged, return to Step 1.

Properties - convergence

- Various strategies can be used to compute the step α^ν. Assume we compute the step to maximize $\theta(\pi^\nu + \alpha r)$ on $\alpha \geq 0$. The uniqueness of the inner minimizer is what makes θ differentiable: an infimum of affine functions of π, it is concave but in general non-smooth, and the residual r is then only a subgradient. Under uniqueness it is the gradient, and the algorithm always produces an ascent direction in θ.
- Stopping on complementarity rather than on $r = 0$ matters for the same reason: at the optimum the inactive constraints keep a strictly negative residual.
- Either the algorithm converges to an optimal solution, or, assuming a bounded set of optimal solutions, it produces an infinite sequence where all limit points are optimal.
- Good performance if the number of dual iterations is small compared to the number of function evaluations that would be required by solving the original problem directly.
- Solving the dual program may be time-consuming, but should nevertheless be easier than solving the original problem directly, as the constraints linking the two stages (i.e. involving x and $y(\xi)$) now appear in the objective.

Another approach: scenarios

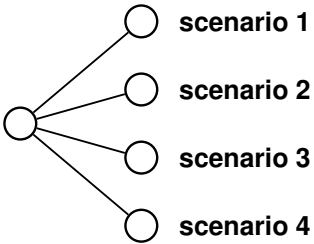
As Ξ is finite, there are S scenarios, and we can link a nonanticipative decision $\hat{x}$ and the decisions of scenario s with the terms $(\hat{x} - x_s)$ in the objective function.

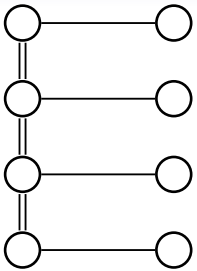
Consider the general multistage problem, over the stages $t = 1, \ldots, H$,

$$\inf_{x \in \mathcal{N}} \mathbb{E}\left[\sum_{t=1}^{H} f_t\left(\boldsymbol{\xi}, x^{t-1}(\boldsymbol{\xi}), x_t(\boldsymbol{\xi})\right)\right],$$
$$\text{s.t. } x_t(\xi) \in X_t(\xi),\ t = 1, \ldots, H, \text{ a.s.},$$

where x_t is the decision at stage t, and $x^t = (x_1, \ldots, x_t)$ the history of the decisions up to stage t (x^0 is empty). $\mathcal{N}$ denotes the closed linear subspace of nonanticipative processes, $\mathcal{X}$ the processes that satisfy the constraints, and $\mathcal{F}$ the σ-field carrying the events, with the probability measure associated to $\boldsymbol{\xi}$.

Nonanticipativity





Mathematically,

- nonanticipativity requires $x^t(\xi)$ to be $\mathcal{F}^t$-measurable, where $\mathcal{F}^t \subseteq \mathcal{F}$ is the σ-field of the events observed up to stage t;
- or, in other words, $x^t(\xi) = \mathbb{E}[x^t(\xi) \,|\, \mathcal{F}^t]$ a.s., $t = 1, \ldots, H$;
- using the projection operator $\Pi^t : z \to \Pi^t z := \mathbb{E}[z \,|\, \mathcal{F}^t]$, this is equivalent to

$$(I - \Pi^t)x^t = 0, \;\; t = 1, \ldots, H.$$

Nonanticipativity (cont'd)

With the scenarios $\mathcal{S} = \{1, \ldots, S\}$, let $\mathcal{S}_t^{(s)}$ be the set of scenarios that share the history $\xi_{[t]} = (\xi_1, \ldots, \xi_t)$ of scenario s, i.e., indistinguishable from s when the decision of stage t is made ($\mathcal{S}_s^t$ in the Multistage slides). Writing $x_t^{(s)}$ the decision of stage t in scenario s, we have to ensure that

$$x_t^{(s)} = x_t^{(s')}, \quad t = 1, \ldots, H, \ s \in \mathcal{S}, \ s' \in \mathcal{S}_t^{(s)}.$$

Idea: explicitly include the nonanticipativity constraints, and place them in the objective function, rather than in the second stage constraints.

Dual ascent and scenarios

Consider the maximization of the dual built on the nonanticipativity constraints:

$$\max_{\pi(\xi)} w = \theta(\pi),$$

where

$$\theta(\pi) = \inf_{x \in \mathcal{X}} \mathbb{E}\left[\sum_{t=1}^{H} f_t\left(\boldsymbol{\xi}, x^{t-1}(\boldsymbol{\xi}), x_t(\boldsymbol{\xi})\right)\right] + \mathbb{E}\left[\sum_{t=1}^{H} \pi^t(\boldsymbol{\xi})^T (I - \Pi^t) x^t(\boldsymbol{\xi})\right],$$

where π^t gathers the multipliers of the constraint $(I - \Pi^t)x^t = 0$, unrestricted in sign since it is an equality.

Dual ascent and scenarios: two stages

We still have to describe the projection operator. For simplicity, consider two stages. The primal problem becomes (omitting all the constraints except the nonanticipativity ones)

$$\min z = \sum_{s=1}^{S} p_s \left(f^1(x_s) + f^2(y_s, \xi_s) \right)$$

$$\text{s.t. } x_s - \sum_{k=1}^{S} p_k x_k = 0, \ s = 1, \ldots, S.$$

The dual problem can now be written as

$$\theta(\pi) = \min_{x,y} \sum_{s=1}^{S} p_s \left(f^1(x_s) + f^2(y_s, \xi_s) + \pi_s^T \left(x_s - \sum_{k=1}^{S} p_k x_k \right) \right).$$

Dual ascent and scenarios: algorithm

Algorithm 2: Lagrangian dual ascent method

Step 0. Let π^0 be arbitrary — unrestricted in sign, the nonanticipativity constraints being equalities — and $\nu = 0$.

Step 1. Given $\pi = \pi^\nu$ in the dual problem, compute the solution $(x_1^\nu, \dots, x_S^\nu, y_1^\nu, \dots, y_S^\nu)$.

Step 2. If $x_s - \sum_{k=1}^S p_k x_k = 0$, $s = 1, \dots, S$, stop: the solution is optimal. Otherwise, define $\hat{\pi}_s = x_s - \sum_{k=1}^S p_k x_k$, and go to Step 3.

Step 3. Let α^ν maximize $\theta(\pi^\nu + \alpha\hat{\pi})$ on $\alpha \geq 0$; no projection is needed here, π being unrestricted. Set $\pi^{\nu+1} = \pi^\nu + \alpha^\nu \hat{\pi}$, $\nu = \nu + 1$. Return to Step 1.

Augmented Lagrangian

- Unfortunately, usually slow as there is a linearization at a single point of θ only. It is nevertheless easy to implement and can give good results, especially for small size problems.
- In order to improve the performances, we will turn to another technique inspired by nonlinear programming: augmented Lagrangian approaches.
- Augmented Lagrangian: add to the Lagrangian a quadratic penalty on the violation of the nonanticipativity constraints.
- Moreover, with augmented Lagrangian techniques, we can try to exploit the problem structure in order to decompose the problem.

Augmented Lagrangian and stochastic programming

We develop the ideas in the two-stage case; the multistage case comes at the end of these slides.

Write the nonanticipativity constraints $x_s = \hat{x}$, $s = 1, \ldots, S$, with a nonanticipative variable $\hat{x}$ in place of $\sum_k p_k x_k$, and add to the Lagrangian a quadratic penalty on their violation:

$$L_\rho(x, y, \hat{x}, \pi) = \sum_{s=1}^{S} p_s \left(f^1(x_s) + f^2(y_s, \xi_s) + \pi_s^T (x_s - \hat{x}) + \frac{\rho}{2} \|x_s - \hat{x}\|^2 \right),$$

where $\rho > 0$ is the penalty parameter, and π_s the vector of multipliers of the nonanticipativity constraints of scenario s.

Augmented Lagrangian and SP: two stages

The augmented Lagrangian dual function is

$$\begin{aligned}\theta_\rho(\pi) = \inf\ & L_\rho(x, y, \hat{x}, \pi) \\ \text{s.t. } & g_i^1(x_s) \le 0,\ i = 1, \dots, m_1,\ s = 1, \dots, S, \\ & b_i^2(x_s, \xi_s) + g_i^2(y_s, \xi_s) \le 0,\ i = 1, \dots, m_2,\ s = 1, \dots, S.\end{aligned}$$

- For fixed $x_1, \dots, x_S$, the minimization in $\hat{x}$ gives $\hat{x} = \sum_s p_s x_s + \frac{1}{\rho} \sum_s p_s \pi_s$: the expected value of the x_s when $\sum_s p_s \pi_s = 0$, a property that the multiplier update of progressive hedging preserves.
- The method aims to contract the pair $(\hat{x}^\nu, \pi^\nu)$ around a saddle point.

Progressive hedging

- Main reference: R. T. Rockafellar and R. J.-B. Wets, *Scenarios and policy aggregation in optimization under uncertainty*, Mathematics of Operations Research 16(1):119–147 (1991).
  The paper treats the multistage case.
- The method performs a complete separation of the problems, scenario by scenario, at each iteration, reducing the cost per iteration.
- The number of iterations can however increase...
- But for structured problems, it is possible to exploit parallelism, and therefore solve large-scale problems.
- Observation: the augmented Lagrangian is not separable with respect to the scenarios: $\hat{x}$ appears in the term of every scenario.

Progressive hedging: basic principle

Alternatively,

1. fix $\hat{x}$ and obtain the solutions x_s, $s = 1, \ldots, S$,
2. fix x_s, $s = 1, \ldots, S$, and compute $\hat{x}$.

In other words, we work scenario by scenario, and we enforce progressively the nonanticipativity constraints.

At iteration ν, we solve the subproblems

$$\begin{aligned} \inf z = \sum_{s=1}^{S} p_s \left(f^1(x_s) + f^2(y_s, \xi_s) + (\pi_s^\nu)^T (x_s - \hat{x}^\nu) + \frac{\rho}{2} \|x_s - \hat{x}^\nu\|^2 \right) \\ \text{s.t. } g_i^1(x_s) \leq 0,\ i = 1, \ldots, m_1,\ s = 1, \ldots, S, \\ b_i^2(x_s, \xi_s) + g_i^2(y_s, \xi_s) \leq 0,\ i = 1, \ldots, m_2,\ s = 1, \ldots, S. \end{aligned}$$

$\hat{x}^\nu$ is not necessarily feasible: each x_s is only feasible for its own scenario.

Decomposition

However, it is easy to decompose the problem!
Algorithm:

Step 0. Choose a nonanticipative $\hat{x}^0$, multipliers π^0 with $\sum_s p_s \pi_s^0 = 0$ (for instance $\pi^0 = 0$), and $\rho > 0$. $\nu \leftarrow 0$. Go to Step 1.

Step 1. Let $\left(x_s^{\nu+1}, y_s^{\nu+1}\right)$ be a solution of the previous Lagrangian program, for $s = 1, \ldots, S$. Set

$$\hat{x}^{\nu+1} = \left(\hat{x}_1^{\nu+1}, \ldots, \hat{x}_S^{\nu+1}\right)^T,$$

$$\text{where } \hat{x}_s^{\nu+1} = \sum_{k=1}^{S} p_k x_k^{\nu+1}, \ s = 1, \ldots, S$$

— the same value for every s, which is exactly what makes $\hat{x}^{\nu+1}$ nonanticipative.

Decomposition (cont'd)

Step 2. Let $\pi_s^{\nu+1} = \pi_s^{\nu} + \rho\left(x_s^{\nu+1} - \hat{x}_s^{\nu+1}\right)$, $s = 1, \ldots, S$.
If $\hat{x}^{\nu+1} = \hat{x}^{\nu}$ and $\pi^{\nu+1} = \pi^{\nu}$, stop: $(\hat{x}^{\nu}, \pi^{\nu})$ is optimal.
Otherwise, set $\nu = \nu + 1$ and return to Step 1.

- Step 1 simply consists of taking the expected value of $x^{\nu+1}$ as $\hat{x}^{\nu+1}$; Step 2 is the multiplier update.
- The basis of this approach is therefore the contraction of the pair $(\hat{x}^{\nu}, \pi^{\nu})$ around a saddle point rather than a dual ascent strategy.
- In practice, the optimality test is replaced by a convergence test for instance if the following quantity is small enough (De Silva and Abramson):

$$\sqrt{\|\hat{x}^{\nu} - \hat{x}^{\nu+1}\|^2 + \sum_{s=1}^{S} p_s \|x_s^{\nu} - \hat{x}^{\nu}\|^2}.$$

Generalization to multistage

$$\min_{x} \sum_{s \in \mathcal{S}} p_s f\left(x^{(s)}, \xi^{(s)}\right)$$

$$\text{s.t. } x^{(s)} \in \mathcal{X}^{(s)},$$

$$x_t^{(s)} \text{ is nonanticipative, } t = 1, \ldots, H.$$

Nonanticipativity:

$$x_t^{(s)} = \mathbb{E}\left[x_t^{(s')} \,\middle|\, s' \in \mathcal{S}_t^{(s)}\right],$$

where $\mathcal{S}_t^{(s)}$ is the set of scenarios that share the history $\xi_{[t]}$ of scenario s. It can be reformulated as

$$x_t^{(s)} = \hat{x}_t^{(s)},$$

with

$$\hat{x}_t^{(s)} = \frac{\sum_{s' \in \mathcal{S}_t^{(s)}} p_{s'} x_t^{(s')}}{\sum_{s' \in \mathcal{S}_t^{(s)}} p_{s'}}.$$

Generalization: augmented Lagrangian

Augmented Lagrangian:

$$L(x,\pi,\rho) = \mathbb{E}\left[f\left(x^{(s)},\xi^{(s)}\right) + \sum_{t=1}^{H}\left({\pi_t^{(s)}}^T \left(x_t^{(s)} - \hat{x}_t^{(s)}\right) + \frac{\rho}{2}\left\| x_t^{(s)} - \hat{x}_t^{(s)} \right\|^2 \right) \right],$$

where π is the vector of Lagrange multipliers of the nonanticipativity constraints, and $\rho > 0$ is a penalty parameter.

Given π, we minimize $L(x,\pi,\rho)$ subject to $x^{(s)} \in \mathcal{X}^{(s)}$, $s \in \mathcal{S}$. To make this program separable by scenario, we fix $\hat{x}_t^{(s)}$, and solve it repeatedly, updating the multipliers and $\hat{x}$ between consecutive resolutions.

Progressive hedging algorithm (PHA)

Step 0. Solve each scenario problem alone, giving $x^{(s),0}$, and set $\hat{x}_t^{(s),0}$ to the averages of Step 2 (next slide). Choose $\pi^{(s),0} = \mathbf{0}$, $\rho^0 > 0$, and set $\nu = 0$.

Step 1. Compute $x^{(s),\nu+1} = \left(x_1^{(s),\nu+1}, \ldots, x_H^{(s),\nu+1}\right)$, $s = 1, \ldots, S$, by solving each scenario subproblem

$$\min_{x^{(s)}} f\left(x^{(s)}, \xi^{(s)}\right) + \sum_{t=1}^{H}\left(\pi_t^{(s),\nu\,T}\left(x_t^{(s)} - \hat{x}_t^{(s),\nu}\right) + \frac{\rho^\nu}{2}\left\|x_t^{(s)} - \hat{x}_t^{(s),\nu}\right\|^2\right)$$

$$\text{s.t. } x^{(s)} \in \mathcal{X}^{(s)}.$$

Progressive hedging algorithm (PHA, cont'd)

Step 2. For $s = 1, \ldots, S$, $t = 1, \ldots, H$, set

$$\hat{x}_t^{(s),\nu+1} = \frac{\sum_{s' \in \mathcal{S}_t^{(s)}} p_{s'} x_t^{(s'),\nu+1}}{\sum_{s' \in \mathcal{S}_t^{(s)}} p_{s'}}.$$

Step 3. Set $\rho^{\nu+1}$ and

$$\pi_t^{(s),\nu+1} = \pi_t^{(s),\nu} + \rho^{\nu} \left(x_t^{(s),\nu+1} - \hat{x}_t^{(s),\nu+1} \right),$$

for $t = 1, \ldots, H$, $s \in \mathcal{S}$.

Step 4. Stop if convergence is achieved. Otherwise, set $\nu \leftarrow \nu + 1$ and return to Step 1.

Convergence

Theorem (Rockafellar and Wets, 1991)

Assume that the stochastic program has convex functions and closed convex constraint sets, that the support of ξ is finite, and that the program has an optimal solution. With a fixed penalty parameter $\rho > 0$, the progressive hedging algorithm converges, from any starting point, to an optimal solution x^* and multipliers π^* forming a saddle point of the Lagrangian; the distance $\|\hat{x}^\nu - x^*\|^2 + \rho^{-2}\|\pi^\nu - \pi^*\|^2$ (norms weighted by the probabilities) decreases at every iteration.

Step 3 lets ρ vary, as many implementations do; the theorem covers a fixed ρ.

- De Silva and Abramson have tested the progressive hedging algorithm for linear problems in portfolio management, proposed by Mulvey and Vladimirou.
- The subproblems were solved by interior point methods.
- Good parallel speed-ups, but the algorithm remains sensitive to the choice of the penalty parameter.

Implementation

The `StochasticPrograms.jl` package proposes an implementation of the progressive hedging algorithm, but limited to two stages.

Some variations or unmaintained versions exist.